

WEEK 4

CONTINGENCY TABLES, CORRELATION & BIVARIATE REGRESSION

APPLIED STATISTICAL ANALYSIS/QUANTITATIVE METHODS I

LECTURER: JEFFREY ZIEGLER, PHD

TEACHING FELLOWS:

ELENA KARAGIANNI

SHEKHAR KEDIA

ASDS - TRINITY COLLEGE DUBLIN

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WHAT IS A CONTINGENCY TABLE?

- A contingency, or “cross-tabulation,” table displays the number of subjects observed at all combinations of possible outcomes for two variables
- Used for analysis of categorical variables
- Ex: Is there an association between gender and party affiliation? Does party affiliation depend on gender?

TABLE 8.1: Party Identification (ID) and Gender, for GSS Data

Gender	Party Identification			Total
	Democrat	Independent	Republican	
Females	573	516	422	1511
Males	386	475	399	1260
Total	959	991	821	2771

WHAT IS A MARGINAL DISTRIBUTION?

- Row totals and column totals are called **marginal distributions**
- For party identification, sample marginal distribution is 959, 991, and 821

Gender	Democrat	Independent	Republican	Total
Females	573	516	422	1511
Males	386	475	399	1260
Total	959	991	821	2771

WHAT IS A JOINT DISTRIBUTION?

Joint distribution: Probability that two variables simultaneously take particular values

What is probability that gender is “female” and party identification is “Democrat”?

$$P(\text{Female and Democrat}) = \frac{573}{2771} = 0.2068 \approx 21\%$$

Gender	Democrat	Independent	Republican	Total
Females	573	516	422	1511
Males	386	475	399	1260
Total	959	991	821	2771

WHAT IS A CONDITIONAL DISTRIBUTION?

Conditional distribution: Probability of one variable taking different values, conditional on another variable having a specific value

What is probability of being a Democrat conditional on being female?

$$P(\text{Democrat} \mid \text{Female}) = \frac{573}{1511} = 0.3792 \approx 38\%$$

Gender	Democrat	Independent	Republican	Total
Females	573	516	422	1511
Males	386	475	399	1260
Total	959	991	821	2771

INDEPENDENCE AND DEPENDENCE

Why is this important? *Agresti & Finlay (2009, 223)*

- Two categorical variables are statistically independent if population conditional distributions on one variable are identical for each category of other
- They are statistically dependent if these conditional distributions differ across categories

TABLE 8.3: Population Cross-Classification Exhibiting Statistical Independence.
The conditional distribution is the same in each row, (44%, 14%, 42%).

Ethnic Group	Party Identification			Total
	Democrat	Independent	Republican	
White	440 (44%)	140 (14%)	420 (42%)	1000 (100%)
Black	44 (44%)	14 (14%)	42 (42%)	100 (100%)
Hispanic	110 (44%)	35 (14%)	105 (42%)	250 (100%)

- Notice:
Conditional distribution is same in each row
- What does that mean?

χ^2 TEST OF INDEPENDENCE

What is χ^2 test of independence?

- We want to test whether two categorical variables are independent

H_O : Variables are statistically independent

H_A : Variables are statistically dependent

Test compares observed and expected frequencies, f_o and f_e :

$$\chi^2 = \sum_{i,j} \frac{(f_{o,ij} - f_{e,ij})^2}{f_{e,ij}}$$

Under independence, the expected frequency is

$$f_{e,ij} = \frac{(\text{row total})(\text{column total})}{N}$$

EXPECTED FREQUENCIES (f_e)

Gender	Democrat	Independent	Republican	Total
Female	573 (522.9)	516 (540.4)	422 (447.7)	1511
Male	386 (436.1)	475 (450.6)	399 (373.3)	1260
Total	959	991	821	2771

For female Democrats,

$$f_e = \frac{959 \times 1511}{2771} = 522.9$$

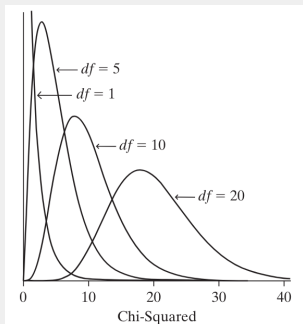
Observed frequency ($f_o = 573$), is somewhat higher than expected ($f_e = 522.9$) under independence

CALCULATING χ^2 TEST STATISTIC

$$\begin{aligned}\chi^2 &= \sum \frac{(f_o - f_e)^2}{f_e} \\ &= \frac{(573 - 522.9)^2}{522.9} + \frac{(516 - 540.4)^2}{540.4} + \dots \\ &= 16.2\end{aligned}$$

- How should we interpret this value?
- How likely are we to observe data in sample (this test statistic), under assumption that H_0 is true?
 - We need to look at χ^2 probability distribution!

THE χ^2 DISTRIBUTION



- Always non-negative
- Right-skewed
- Shape depends on degrees of freedom
 - ▶ Mean: $\mu = df$
 - ▶ Standard deviation: $\sigma = \sqrt{2df}$
 - ▶ As $df \uparrow$, skew $\downarrow$ and curve becomes more bell-shaped

For an $r \times c$ contingency table,

$$df = (r - 1)(c - 1)$$

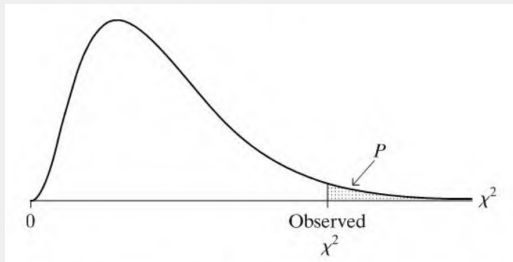
For a 2×3 table,

$$df = (2 - 1)(3 - 1) = 2$$

Degrees of freedom determine exact shape of χ^2 distribution

CALCULATE p -VALUE

- Larger χ^2 values provide stronger evidence against independence
- p -value is right-tail probability under H_0
- It's probability of obtaining a test statistic at least as extreme as observed value



Decision rule

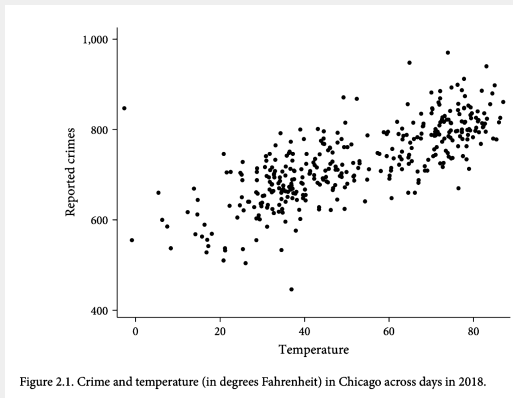
If $p < \alpha$, for example 0.05, reject H_0

SCATTERPLOTS

Scatterplot shows two continuous variables alongside each other

For each observation, value of X is plotted against value of Y

- “each point corresponds to an observation in our data—here, that means each point is a day in Chicago in 2018”
- Point’s location shows average temperature (X) and crime (Y)



Bueno de Mesquita and Fowler (2021, 15–16)

WHAT IS CORRELATION?

Correlation describes the extent to which two features of the world tend to occur together

- Positive correlation: $\uparrow$ values of one feature tend to occur with $\uparrow$ values of other
- No correlation: No linear systematic pattern occurs together
- Negative correlation: $\uparrow$ values of one feature tend to occur with $\downarrow$ values of other

HOW CAN WE MEASURE CORRELATION?

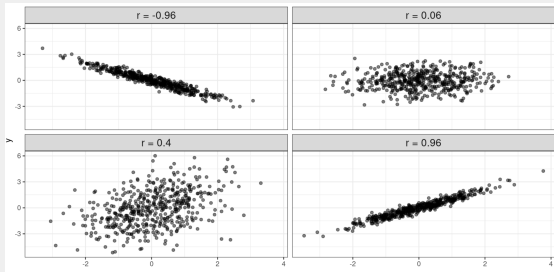
Most common measure is Pearson correlation coefficient r

- Standardized covariance between two variables X and Y

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

- $r \in [-1, 1]$
- $r = 0$: No linear association
- $r > 0$: + association
- $r < 0$: - association
- Closer $|r|$ is to 1 = stronger association

SHORTCOMINGS OF CORRELATION ANALYSIS



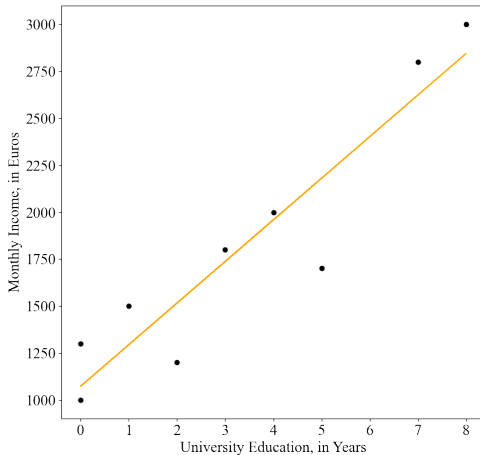
- No indication on “substantive importance or size of relationship between X and Y”
- Slope **only** “tells us, descriptively, how much Y changes, on average, as X increases by one unit”

Bueno de Mesquita and Fowler (2021, 29)

WHAT IS A LINEAR REGRESSION MODEL?

Find line with "best fit":

$$Y_i = \alpha + \beta X_i + \epsilon_i$$



ORDINARY LEAST SQUARES (OLS)

- Best line minimises sum of squared errors (SSE)
- Residuals are vertical deviations from line:

$$e_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i) = y_i - \hat{y}_i$$

- SSE is sum of squared differences between actual and predicted values:

$$SSE = \sum_{i=1}^n \hat{\epsilon}_i^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

- Goal is to minimize this!

THINGS WE NEED TO ASSUME FOR THIS TO WORK

Suppose we have bivariate linear model:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Two assumptions are sufficient to derive unbiased OLS coefficients:

1. Parameters in model represent **linear** associations between X and Y
2. Zero conditional mean:

$$E(\epsilon_i | X) = 0$$

ADDITIONAL ASSUMPTIONS

To estimate variance of coefficients, we also assume:

- Constant conditional variance, or homoskedasticity:

$$\text{Var}(\epsilon_i | X) = \sigma^2$$

- No correlation between error terms:

$$\text{Cov}(\epsilon_i, \epsilon_j | X_i, X_j) = 0, \quad i \neq j$$

- Normally distributed errors:

$$\epsilon_i | X \sim N(0, \sigma^2)$$

STANDARD ERRORS FOR REGRESSION COEFFICIENTS

If zero conditional mean assumption holds ($(\epsilon_i | X) = 0$),

$$E(\hat{\beta}_0) = \beta_0, \quad E(\hat{\beta}_1) = \beta_1$$

Assuming normally distributed errors, OLS coefficients are normally distributed:

$$\hat{\beta}_0 \sim N\left(\beta_0, \text{Var}(\hat{\beta}_0)\right),$$

$$\hat{\beta}_1 \sim N\left(\beta_1, \text{Var}(\hat{\beta}_1)\right)$$

This allows us to calculate **standard errors**

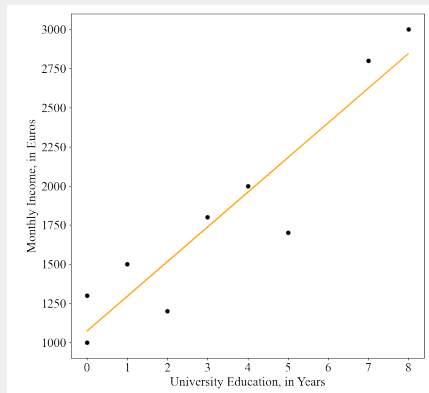
INTERPRETING REGRESSION MODEL

$$\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i$$

- $\hat{Y}_i$: predicted outcome
- α : intercept, expected value of Y when $X = 0$
- β : slope, expected change in Y for a one-unit increase in X
- ϵ_i : error or residual, $\epsilon_i = Y_i - \hat{Y}_i$

INTERPRETING ESTIMATED MODEL

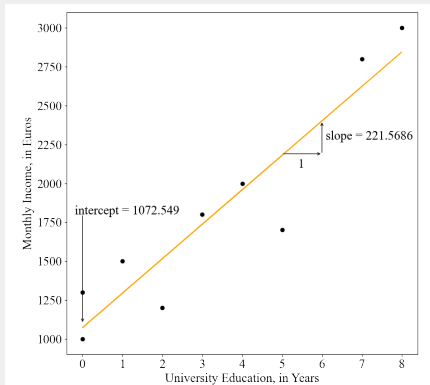
Estimated regression:



$$\widehat{\text{income}} = \hat{\alpha} + \hat{\beta} \text{ education}$$

$$\widehat{\text{income}} = 1072.55 + 221.57 \text{ education}$$

INTERPRETING COEFFICIENTS



Intercept: $\hat{\alpha} = 1072.549$

Expected income when education equals zero years:

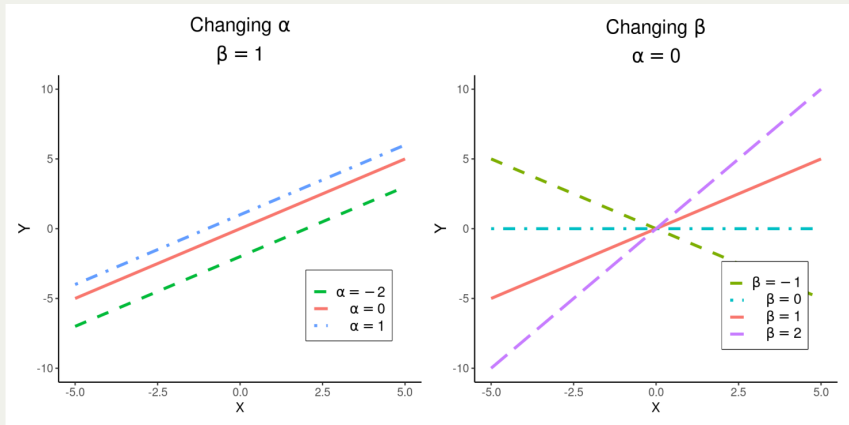
$$1072.549 + 221.5685(0) = 1072.549$$

Slope: $\hat{\beta} = 221.5685$

Each additional year of university education increases expected income by 221.57 euros, on average:

$$1072.549 + 221.5685(1) = 1294.1175$$

VARIETIES OF LINEAR RELATIONSHIPS (α & β)

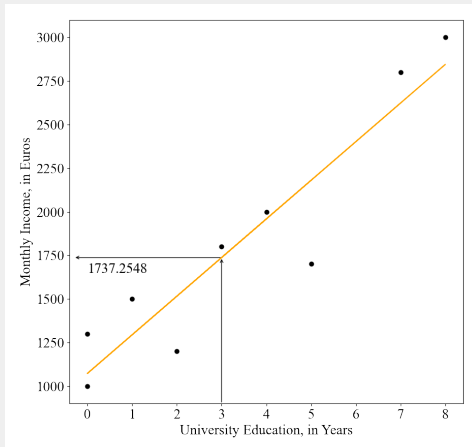


MAKING PREDICTIONS

For education equal to 3 years:

$$\widehat{\text{income}} = 1072.55 + 221.57(3) = 1737.25$$

We can plug any value of X , measured in years of education, into model to predict expected income



RESIDUALS

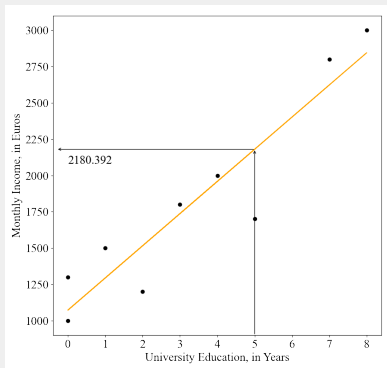
For a person with 5 years of education:

$$\widehat{\text{income}} = 1072.55 + 221.57(5) = 2180.39$$

If actual income is 1700 euros:

$$\text{Residual} = Y - \hat{Y} = 1700 - 2180.39 = -480.39$$

A negative residual means that observed income is below predicted income



CLASS BUSINESS

- Read required (and suggested) online materials
- Finish Problem set #1!